“Best in PA” Universities & Colleges Category Classification Ensemble Model

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Description of the Project

In the project development, the goal was to provide students with a model that can accurately classify a Pennsylvanian post-secondary institution category for them based on geographical data, an estimate of tuition cost, institution name, and region. This project uses an ensemble multi-class classification model that classifies Pennsylvania universities', colleges', and other institutions' categories based on their data, and it uses a real-world modified dataset that can provide accurate metrics with estimations of missing data.

Significance of the Project

This project can be for research purposes or for high school users making decisions about post-secondary institutions. Based on a specific institution input, it can be classified as the category the institution is a part of, thus the type. This allows for researchers to classify institution data based on the categorical label it has, and it allows for high schools to see how a specific institution categorizes and if it is the type of institution they are looking for.

Data Collection

Through the data research process, the team analyzed Pennsylvania government open-access database to find suitable data for institution prediction. A limitation during dataset research was that there was not tuition data available for all institutions. We utilized government databases to ensure data accuracy and accessibility, thus high quality. The data was collected by downloading CSV files and reading them in with the Pandas data processing library. From here, we performed lots of filtering, merging, and imputation, and the dataset reduced to 1,200 rows of post-secondary institutions within Pennsylvania. The following figures show the first 5 lines of each of the three data files (unprocessed) and the first 5 lines of the preprocessed dataset:

```
                      Sector                          College  \
0  Private College/University                    Wilson College
1  Private College/University                 Wilkes University
2  Private College/University      York College of Pennsylvania
3  Private College/University  Won Institute of Graduate Studies
4  Private College/University                   Ursinus College

                  Type      County  zip code  \
0  Main Campus Location    Franklin     17201
1  Main Campus Location     Luzerne     18766
2  Main Campus Location        York     17403
3  Main Campus Location  Montgomery     18974
4  Main Campus Location  Montgomery     19426

                                       Address   Latitude  Longitude  \
0  1015 Philadelphia Ave, Chambersburg, PA 17201  39.947000 -77.653737
1          84 W South St, Wilkes-Barre, PA 18766  41.244365 -75.889973
2            441 Country Club Rd, York, PA 17403  39.947447 -76.726798
3      800 Jacksonville Rd, Warminster, PA 18974  40.202613 -75.076139
4          601 E Main St, Collegeville, PA 19426  40.191770 -75.456848

  Georeferenced Latitude & Longitude                          Sector and Type
0     POINT (-77.6537371 39.9470004)  Private College/University - Main Campus
1     POINT (-75.8899735 41.2443653)  Private College/University - Main Campus
2     POINT (-76.7267984 39.9474471)  Private College/University - Main Campus
3       POINT (-75.076139 40.202613)  Private College/University - Main Campus
4     POINT (-75.4568484 40.1917705)  Private College/University - Main Campus
```

Figure 1. Higher Education Institutions by Sector Dataset Sample.

```
                          Institution Name  Administrative Unit Number  \
0              AAA School of Trucking, Inc                   426510819
1             AAA School of Trucking, Inc.                   415228149
2       Academy for Media Production (The)                   412015229
3  Academy of Investigation and Protection                   427041279
4   Academy of Massage Therapy & Bodyworks                   423461299

   School Branch                 Category Status Phone Number  \
0              0  Private Licensed School   Open   2673245957
1              0  Private Licensed School   Open   7176523652
2              0  Private Licensed School   Open   7173344487
3              0  Private Licensed School   Open   7247757970
4              0  Private Licensed School   Open   6104685497

   Phone_Number_Extn                          Web_Address        County  \
0                NaN   http://www.aaaschooloftrucking.com  Philadelphia
1                NaN  http://www.aaaschooloftrucking.com/       Dauphin
2                NaN              https://amp-school.com/         Adams
3                NaN                                  NaN        Beaver
4                NaN     http://www.massage-training.net/    Montgomery

            City    Primary Administrator School Year
0   Philadelphia         Mr Ryan A Soloby      SY2021
1     Harrisburg       Ms Susan     First      SY2021
2  McSherrystown  Mr Paul T Chiacchierini      SY2021
3         Monaca       Mr Louis W Gentile      SY2021
4      Pottstown    Ms Angela  Pio Wagner      SY2021
```

Figure 2. Higher Education Institutions Dataset Sample.

```
   UnitID                    Institution Name  \
0  210508                Academy of Vocal Arts
1  210571                    Albright College
2  210599             All-State Career School
3  418658  All-State Career School-Pittsburgh
4  210669                   Allegheny College

   Total price for in-district students living on campus  2024-25 (DRVCOST2024)  \
0                                                    NaN
1                                                47508.0
2                                                    NaN
3                                                    NaN
4                                                74150.0

   Total price for in-state students living on campus 2024-25 (DRVCOST2024)  \
0                                                    NaN
1                                                47508.0
2                                                    NaN
3                                                    NaN
4                                                74150.0

   Total price for in-district students living off campus (not with family)  2024-25 (DRVCOST2024)  \
0                                                    NaN
1                                                47583.0
2                                                    NaN
3                                                    NaN
4                                                71346.0

   Total price for in-state students living off campus (not with family)  2024-25 (DRVCOST2024)  \
0                                                    NaN
1                                                47583.0
2                                                    NaN
```

```
   Total price for in-state students living off campus (not with family)  2024-25 (DRVCOST2024)  \
0                                                    NaN
1                                                47583.0
2                                                    NaN
3                                                    NaN
4                                                71346.0

   Total price for in-district students living off campus (with family)  2024-25 (DRVCOST2024)  \
0                                                    NaN
1                                                33156.0
2                                                    NaN
3                                                    NaN
4                                                65672.0

   Total price for in-state students living off campus (with family)  2024-25 (DRVCOST2024)  \
0                                                    NaN
1                                                33156.0
2                                                    NaN
3                                                    NaN
4                                                65672.0

  State abbreviation (HD2024)  Unnamed: 9
0                          PA         NaN
1                          PA         NaN
2                          PA         NaN
3                          PA         NaN
4                          PA         NaN
```

Figure 3-4. Higher Education Institutions by Tuition Sample.

```
Train X:              x0        x8       x10       x12       x13       x14       x15  \
0     0.756659 -0.016682 -0.016682 -0.016682 -0.510567  0.780891  0.796833
1     0.756659 -0.016682 -0.016682 -0.016682 -0.510567  0.780891  0.796833
2    -1.232703 -0.016682 -0.016682 -0.016682  1.958605 -1.280589 -1.254968
3    -0.213450 -0.016682 -0.016682 -0.016682 -0.510567 -1.280589  0.796833
4    -0.213450 -0.016682 -0.016682 -0.016682 -0.510567 -1.280589  0.796833
5    -0.979830 -0.016682 -0.016682 -0.016682 -0.510567 -1.280589 -1.254968
6    -3.249197 -4.120388 -4.120388 -4.120388 -0.510567 -1.280589  0.796833
7     0.756659 -0.016682 -0.016682 -0.016682 -0.510567  0.780891  0.796833
8    -1.232703 -0.016682 -0.016682 -0.016682  1.958605 -1.280589 -1.254968
```

Figure 5. Preprocessed Dataset (Train X) as Model Input Sample.

```
Test X:             x0        x8       x10       x12       x13       x14       x15  \
0   -0.213450 -0.016682 -0.016682 -0.016682 -0.510567 -1.280589  0.796833
1   -0.591326 -0.016682 -0.016682 -0.016682  1.958605  0.780891 -1.254968
2    0.756659 -0.016682 -0.016682 -0.016682 -0.510567  0.780891  0.796833
3   -0.591326 -0.016682 -0.016682 -0.016682  1.958605  0.780891 -1.254968
4   -1.308562 -0.016682 -0.016682 -0.016682 -0.510567 -1.280589 -1.254968
5    0.756659 -0.016682 -0.016682 -0.016682 -0.510567  0.780891  0.796833
6   -0.213450 -0.016682 -0.016682 -0.016682 -0.510567 -1.280589  0.796833
7   -0.213450 -0.016682 -0.016682 -0.016682 -0.510567 -1.280589  0.796833
8   -1.232703 -0.016682 -0.016682 -0.016682  1.958605 -1.280589 -1.254968
```

Figure 6. Preprocessed Dataset (Test X) as Model Input Sample.

```
Train Y:  922             Private College and University
7065            Private College and University
624             Private College and University
915      State-Related Commonwealth University
5981     State-Related Commonwealth University
8101            Private College and University
31                  Private College/University
5986     State-Related Commonwealth University
9006                      Other Post Secondary
33                           Community College
3931            Private College and University
9824           Private State-Aided Institution
```

Figure 7. Preprocessed Dataset (Train Y) as Model Input Sample.

```
Test Y:  9398            Private College and University
9138     State-Related Commonwealth University
774      State-Related Commonwealth University
2801     State-Related Commonwealth University
1948                          State University
295      State-Related Commonwealth University
6947     State-Related Commonwealth University
6945     State-Related Commonwealth University
3467            Private College and University
8081            Private College and University
9135                  Private Two-Year College
```

Figure 8. Preprocessed Dataset (Test Y) as Model Input Sample.

Data Processing and Feature Engineering

To preprocess the data, we used typical data preprocessing steps. We began by reading in the three CSV files of the data. From here, each of these three DataFrames had some of the columns renamed, so when they were concatenated into 1 DataFrame, the columns merged into the same column (avoid having 3 columns with the same names). We also dropped columns that were clearly not useful towards the data (such as phone numbers, web addresses, etc.). Then, the three DataFrames were merged. Once they were merged, the categories (target variable) were filtered. Some of the datasets included categories such as administrative offices, technical programs, and others, and these were removed. The goal of this project is to simply determine between university, college, and degree program types.

From here, we moved towards feature engineering. We created fifteen new features using the geographical data. Using the longitude and latitude column of the dataset, we found the geographical longitude and latitude of the top five largest cities in Pennsylvania. Using this information, we found the geographical distance from every institution to the each of the cities, creating ten new features. From here, we performed some manual feature engineering. By

finding all unique counties in the dataset, we manually sorted these counties into 5 region categories (North, East, South, West, and Central), allowing for a county to fit into multiple regions. These 5 columns are binary, so 1 was used for the county being part of the region and 0 was used for not being part of the region. It is important to note that feature engineering was performed before imputation because the feature engineered columns also required imputation.

After feature engineering, we created imputers, both numerical and categorical. The numerical imputer was an Iterative Imputer from the Scikit-Learn library, which iterates through the data multiple times to find the best imputative value, making it more robust for our purposes since the dataset needed much imputation. For the categorical columns, we used a Simple Imputer that used the most frequent value since the Iterative Imputer is not applicable to categorical data.

After imputation, we performed encoding. Based on our dataset, it was recommended to encode before the train test split of the data because we had an error of encoding after the split, due to unseen values in the train data within the testing data. Therefore, we could not implement encoding after since the encoder could not encode the unseen values. From here, the data was split into 90% training data and 10% testing data. We used a smaller testing sample since we had limited training data. From here, the data was scaled using a Standard Scaler process. After scaling, we performed feature selection. To perform feature selection, we selected the top 15 features from the dataset, and we passed this final dataset to the models.

Model Development

As input, the model received the top 15 features that we found during data preprocessing (around half of the features of the dataset). After using these features as input, the model's goal

was to categorize the institution by understanding the type of institution a data iteration may be based on the data.

To develop our model, we created six different classifiers that we have learned during the duration course. We combined these six classifiers into an ensemble classification model by creating a soft voting ensemble classifier by using pipelines.

Firstly, we created a K-Nearest Neighbors model. This model is useful for our specific uses because the data has many imputations, and similar data can be grouped into similar categories based on the information available. We implemented this with the following hyperparameters for the model: 20 neighbors, weights by distance, and the brute algorithm. We used more neighbors to ensure the model was getting enough to data to classify from the surround data, and we used distance for the weights instead of uniform to get a greater robustness on the model, ensuring that the calculated neighbor was close in distance based on the weight. We used a brute algorithm to ensure all solutions were checked and met.

Secondly, we created a Decision Tree Classifier model. This model was used as it is a simple but effective tree mechanism for classification, allowing for detailed customization. We chose a log loss function to allow a minimization to the errors. We chose a max depth of the trees as 10 to prevent the model from overfitting, and by default, we used the best splitter option.

Thirdly, we created a Random Forest Classifier model, which was one of our best models. This model is a powerful classification model that utilizes bagging on the data to parallel train its internal models. We utilized 200 estimators instead of the default 100 to allow for more parallelization, and we used a log loss function again to allow for minimization of the errors in the function. We add a max features at each split as 6 which is a bit over a third of all the features

to ensure that this model is only using the best features for the given split, furthering narrowing down to the best data, and balanced the class weight (according to our sources, this is a crucial parameter for imbalanced data.

Fourthly, we created a Logistic Regression Classifier model. This model is another simple model that works well with classification tasks. We chose a low C value, so that the model has room to make errors and is not penalized as heavily. We chose the Stochastic Average Gradient (SAG) solver which essentially works well when there is many features. We used a max iteration amount of 1000 which is recommended for finding the optimal solution, necessary for our complex dataset and model.

Fifthly, we created a Gradient Boosting Classifier model, and this model performs exceptionally with a complex dataset, low performing models, improving performance internally, and learning from errors and the gradient. We used 300 estimators for maximum model solution optimality. We lowered the learning rate from the default 0.1 to 0.05 so that the model can learn more details of the data and fit well to it.

Sixthly, we created an Adaptive Boosting Classifier model, and this model does well with complex data and it adapts to the data and understands how to improve errors. We used 300 estimators and a 0.05 learning rate for this model as well for the same reason as the Gradient Boosting Classifier model.

Finally, in our initial testing, we created a Support Vector Machine Classifier model, but we found that this model is not suitable for the task we needed. We chose to not implement this model as we already had 6 models going into our ensemble.

The following figures show the test results of executing our models as 1 Soft Voting Classifier Ensemble Model:

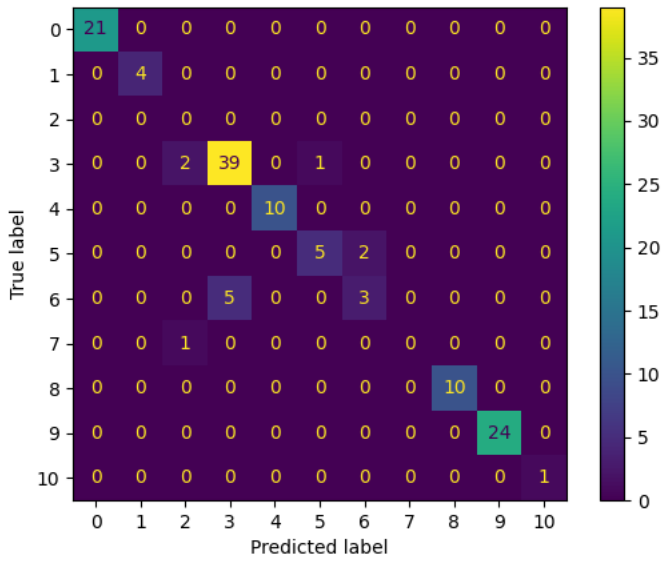

Figure 9. Confusion Matrix.

```
KNN Score:  0.8515625
DTC Score:  0.8125
RFC Score:  0.9140625
LRC Score:  0.546875
GBC Score:  0.9140625
ADA Score:  0.4296875
Ensemble Score:  0.9140625
```

Figure 10. Model Accuracy Scores in Terminal.

	precision	recall	f1-score	support
Community College	1.00	1.00	1.00	21
Other College and University	1.00	1.00	1.00	4
Other Post Secondary	0.00	0.00	0.00	0
Private College and University	0.89	0.93	0.91	42
Private College/University	1.00	1.00	1.00	10
Private State-Aided Institution	0.83	0.71	0.77	7
Private Two-Year College	0.60	0.38	0.46	8
Rural Regional College	0.00	0.00	0.00	1
State University	1.00	1.00	1.00	10
State-Related Commonwealth University	1.00	1.00	1.00	24
Theological Seminary	1.00	1.00	1.00	1
accuracy			0.91	128
macro avg	0.76	0.73	0.74	128
weighted avg	0.92	0.91	0.92	128

Figure 11. Classification Report in Terminal.

Showcase

The trained model is used in our application by allowing for training and testing onto the dataset and displaying classification performance metrics and plots. When the application is run, the user can select train and then test the model lineup. To make the classifier accessible beyond our local laptops, we first transitioned the executable app to a web-based deployment using Flask. This allowed us to preserve the existing ML logic in data.py and models.py without modification.

Furthermore, the backends' endpoint GET /stream/<job_id> route uses Server-Sent Events (SSE) to push real-time stdout output from the training process back to the browser. For

hosting, we used Railway as a deployment platform due to its easy GitHub integration. Upon connecting the repository, Railway detected the Flack application and deployed it on each new commit. We added Procfile to the repository root to instruct Railway how to start the server. Lastly, a requirements text file was included to make sure all Python dependencies were installed in the deployment environment.

To access the models training and testing without a user interface, access the following GitHub link: https://github.com/daisyapto/CMPSC445_Project2. To access the code for the deployed application, access the following GitHub link: https://github.com/Zainabzz/COMPSC-445_FINAL. The following figures show snapshots of the UI in action:

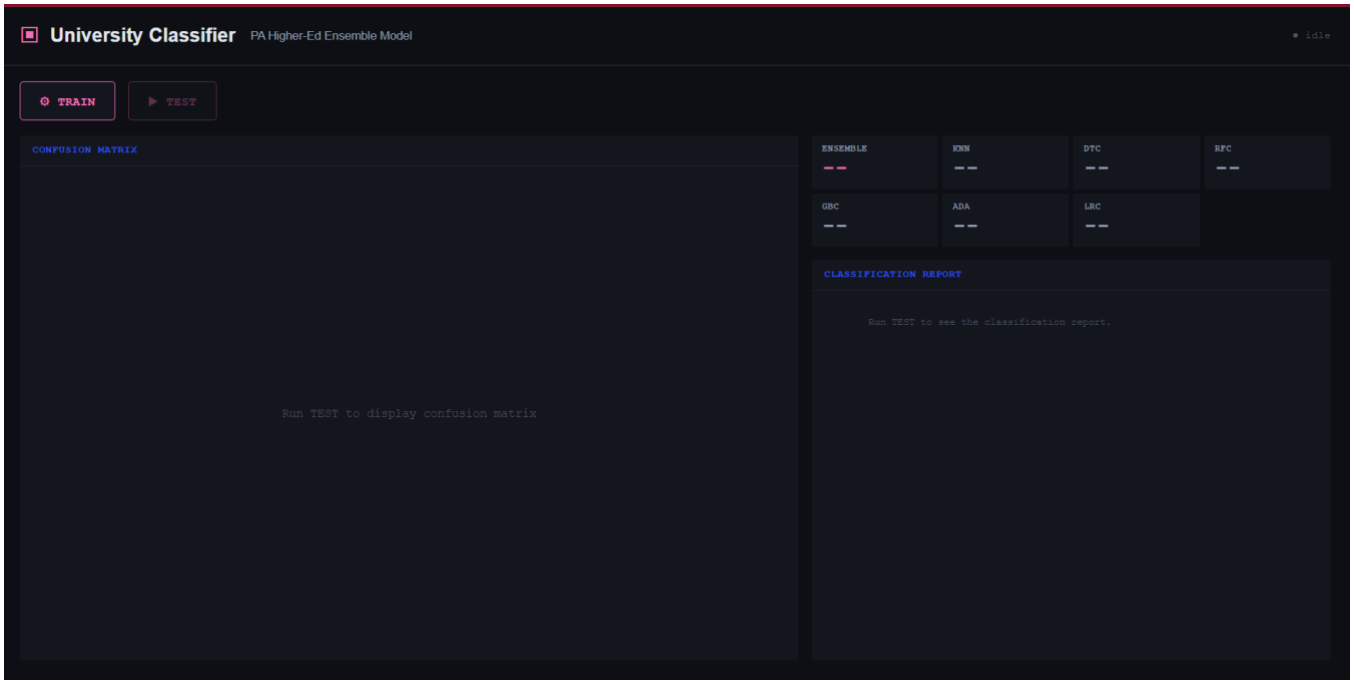

Figure 12a. User Interface Start Page.

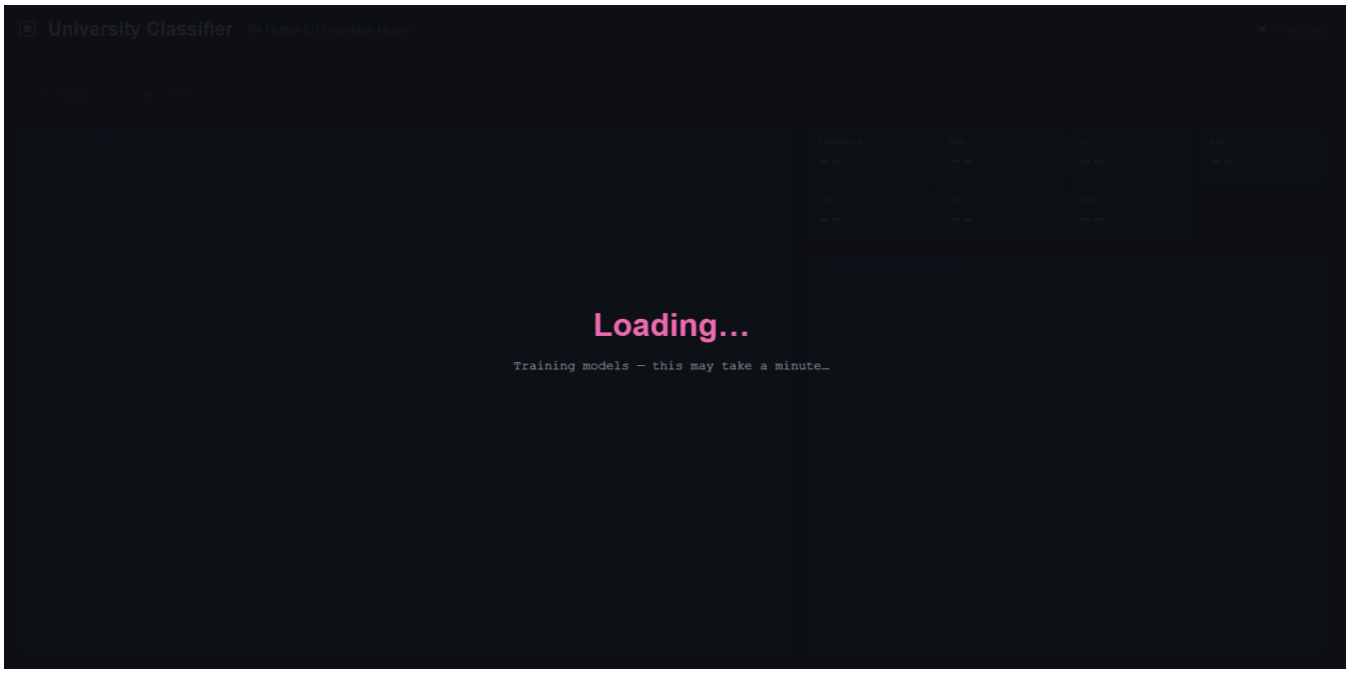

Figure 12b. User Interface Loading Page.

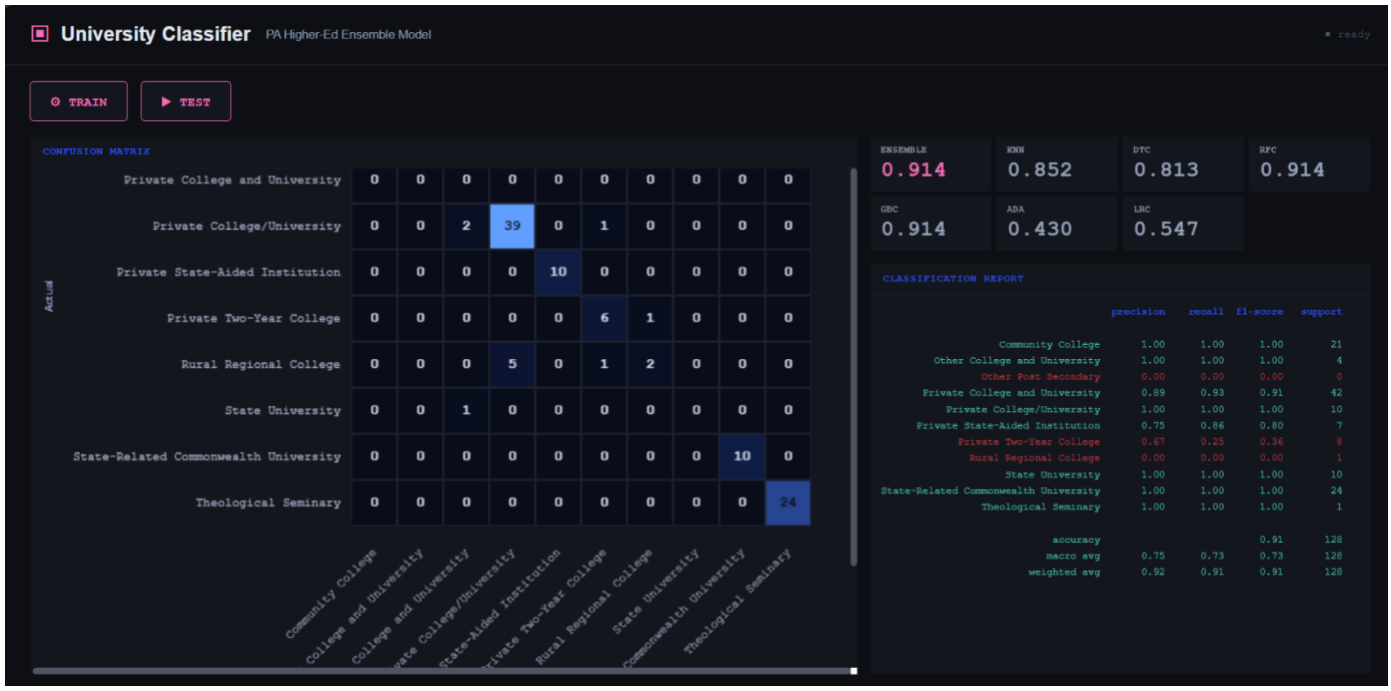

Figure 12c. User Interface Final.

Discussion and Conclusions

During development, there were some limitations. Firstly, the dataset included missing values since the data was merged from multiple sources, resulting in empty values. We combated this by preforming feature engineering, data filtering, and imputation. Secondly, through the model development, we found that predicting the institution name resulted in low accuracy since there was only one iteration of each institution, making it quite difficult for the models to accurately predict a specific institution based on many features. We shifted the focus to the prediction of the category of institution which created much better performance for the model, providing more accurate and thus useful predictions.

In the development of the UI, there were a few connectivity issues that arose. Integrating the data portion to the front-end was a multi-step process and solving issues that arose required going back multiple steps. Firstly, sizing the different windows onto the website took time due to

text and graph placement attempts. It took multiple attempts to proportion the size correctly and make it wrap so that it changes based on the device its opened on. Secondly, we faced many “not found” errors, which occurred because Flask required HTML templates to live inside folders with specific names.

To conclude, this project utilizes real-world data for classification on the best university category for an institution. This project combines the performance of multiple models to provide one accurate result. Through this course and project, we synthesized our knowledge on many machine learning models, and we combined them into one functional, practical application.

AI Usage

In this project, Daisy Aptovska utilized AI for two purposes: data research and debugging. Gemini was utilized to ask questions on how to find datasets for this tasks and useful database websites to consult. Google search AI Overview was utilized for searching how to resolve error messages, suggestions for improving model performance, and questions surrounding the imputation limitation of the dataset.

Zainab Afridi utilized AI by assisting with UI debugging and code resource package finding. She used Claude AI to assist with deployment steps and resources. Claude helped solve errors that came up while deploying the website VIA flask.

References

https://data.pa.gov/Post-Secondary-Education/PA-Higher-Education-Institutions-by-Sector-Current/8u5h-xvus/about_data

https://catalog.data.gov/dataset?q=pennsylvania

https://github.com/daisyapto/CMPSC445_Project1

https://www.pasda.psu.edu/

https://catalog.data.gov/dataset/?q=resturants&sort=views_recent+desc&ext_location=&ext_bbox=&ext_prev_extent=

https://business.yelp.com/data/resources/open-dataset/

https://data.pa.gov/browse?sortBy=relevance&page=1&pageSize=20

https://data.pa.gov/Post-Secondary-Education/Supply-Demand-Gap-Analysis-2014-2024-State-System-/n3sx-ckrr/about_data

https://en.wikipedia.org/wiki/List_of_counties_in_Pennsylvania

https://scikit-learn.org/stable/modules/ensemble.html

https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.StackingClassifier.html

https://scikit-learn.org/stable/modules/generated/sklearn.impute.SimpleImputer.html

https://scikit-learn.org/stable/modules/generated/sklearn.impute.KNNImputer.html

https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html#sklearn.neighbors.KNeighborsClassifier

https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html

https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.classification_report.html

https://scikit-learn.org/stable/modules/impute.html

https://scikit-learn.org/stable/modules/generated/sklearn.impute.IterativeImputer.html

https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.SelectKBest.html#sklearn.feature_selection.SelectKBest

https://colab.research.google.com/drive/17gnNWQZh33lMmOdcHpeg6bP78hJsjtVw?usp=sharing

https://www.collegetuitioncompare.com/compare/tables/?state=PA